

Corrected Derivation of the Quasi-Linearized Cole-Hopf Transformation for Viscous Hamilton-Jacobi PDEs

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Abstract

We provide a corrected and complete derivation of the Cole-Hopf-like transformation proposed for linearizing the viscous Hamilton-Jacobi (HJ) PDE. We show that the transformation is *exact* only when the Hamiltonian is quadratic in the co-state ($H = \frac{1}{2}|\mathbf{p}|^2$), and derive the precise residual that arises for general Hamiltonians. We then formulate the quasi-linearization iteration and establish the Gaussian expectation formulas for the value function and its spatial gradient.

1 Setup

Consider the viscous HJ initial-value problem:

$$\mathbf{v}_t^\delta + H(t; \mathbf{x}, D\mathbf{v}^\delta) = \frac{\delta}{2} \Delta \mathbf{v}^\delta \quad \text{in } \mathbb{R}^n \times (0, T], \quad \mathbf{v}^\delta(0; \mathbf{x}) = g(\mathbf{x}), \quad (1)$$

where $\delta > 0$ is the viscosity parameter, $H : \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ is the Hamiltonian, and $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is the terminal/initial cost.

2 The Cole-Hopf Transformation

2.1 Exact case: $H(\mathbf{p}) = \frac{1}{2}|\mathbf{p}|^2$

Define $\omega^\delta = e^{-\mathbf{v}^\delta/\delta}$. Then:

$$\omega_t^\delta = -\frac{1}{\delta} \omega^\delta \mathbf{v}_t^\delta, \quad (2)$$

$$D\omega^\delta = -\frac{1}{\delta} \omega^\delta D\mathbf{v}^\delta, \quad (3)$$

$$\Delta \omega^\delta = \frac{1}{\delta^2} \omega^\delta |D\mathbf{v}^\delta|^2 - \frac{1}{\delta} \omega^\delta \Delta \mathbf{v}^\delta. \quad (4)$$

From (1) with $H = \frac{1}{2}|D\mathbf{v}^\delta|^2$:

$$\mathbf{v}_t^\delta = \frac{\delta}{2} \Delta \mathbf{v}^\delta - \frac{1}{2} |D\mathbf{v}^\delta|^2.$$

Substituting:

$$\omega_t^\delta = -\frac{1}{\delta} \omega^\delta \left(\frac{\delta}{2} \Delta \mathbf{v}^\delta - \frac{1}{2} |D\mathbf{v}^\delta|^2 \right) = -\frac{1}{2} \omega^\delta \Delta \mathbf{v}^\delta + \frac{1}{2\delta} \omega^\delta |D\mathbf{v}^\delta|^2, \quad (5)$$

$$\frac{\delta}{2} \Delta \omega^\delta = \frac{1}{2\delta} \omega^\delta |D\mathbf{v}^\delta|^2 - \frac{1}{2} \omega^\delta \Delta \mathbf{v}^\delta. \quad (6)$$

Therefore $\omega_t^\delta = \frac{\delta}{2} \Delta \omega^\delta$, i.e., ω^δ satisfies the **heat equation exactly**.

Proposition 1 (Exact Cole-Hopf). *If $H(\mathbf{p}) = \frac{1}{2}|\mathbf{p}|^2$, then $\omega^\delta = e^{-\mathbf{v}^\delta/\delta}$ satisfies*

$$\omega_t^\delta = \frac{\delta}{2} \Delta \omega^\delta \quad \text{in } \mathbb{R}^n \times (0, T], \quad \omega^\delta(0; \mathbf{x}) = e^{-g(\mathbf{x})/\delta}.$$

2.2 General case: the residual

For a general Hamiltonian, define $c(t, \mathbf{x}) := \frac{2H(t; \mathbf{x}, D\mathbf{v}^\delta)}{\delta |D\mathbf{v}^\delta|^2}$ and set $\omega^\delta = e^{-c\mathbf{v}^\delta}$. Writing $f = -c\mathbf{v}^\delta$, the chain rule gives:

$$\omega_t^\delta = \omega^\delta (-c_t \mathbf{v}^\delta - c \mathbf{v}_t^\delta), \quad (7)$$

$$\Delta \omega^\delta = \omega^\delta (|Df|^2 + \Delta f), \quad (8)$$

where $Df = -Dc \mathbf{v}^\delta - c D\mathbf{v}^\delta$ and $\Delta f = -\Delta c \mathbf{v}^\delta - 2\langle Dc, D\mathbf{v}^\delta \rangle - c \Delta \mathbf{v}^\delta$.

Derivation of (8). We compute $\Delta \omega^\delta$ for $\omega^\delta = e^f$ where $f(\mathbf{x}) = -c\mathbf{v}^\delta$.

Step 1 (Gradient of ω^δ). By the chain rule applied to $\omega^\delta = e^f$:

$$\frac{\partial \omega^\delta}{\partial x_i} = e^f \frac{\partial f}{\partial x_i}, \quad \text{i.e.,} \quad D\omega^\delta = \omega^\delta Df.$$

Step 2 (Second partial derivatives). Differentiate $\partial_{x_i} \omega^\delta = e^f \partial_{x_i} f$ once more using the product rule:

$$\frac{\partial^2 \omega^\delta}{\partial x_i^2} = \frac{\partial}{\partial x_i} \left(e^f \frac{\partial f}{\partial x_i} \right) = \underbrace{e^f \left(\frac{\partial f}{\partial x_i} \right)^2}_{\text{chain rule on } e^f} + \underbrace{e^f \frac{\partial^2 f}{\partial x_i^2}}_{\text{product rule remainder}}.$$

Step 3 (Sum over all dimensions). The Laplacian $\Delta \omega^\delta = \sum_{i=1}^n \partial_{x_i}^2 \omega^\delta$ gives:

$$\Delta \omega^\delta = \sum_{i=1}^n e^f \left[\left(\frac{\partial f}{\partial x_i} \right)^2 + \frac{\partial^2 f}{\partial x_i^2} \right] = e^f \left[\underbrace{\sum_{i=1}^n \left(\frac{\partial f}{\partial x_i} \right)^2}_{=|Df|^2} + \underbrace{\sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2}}_{=\Delta f} \right] = \omega^\delta (|Df|^2 + \Delta f).$$

Step 4 (Expand Df and Δf via the product rule). Since $f = -c\mathbf{v}^\delta$ and $c = c(t, \mathbf{x})$ is itself spatially varying:

$$\begin{aligned} Df &= -Dc \mathbf{v}^\delta - c D\mathbf{v}^\delta, \\ \Delta f &= -\Delta c \mathbf{v}^\delta - 2\langle Dc, D\mathbf{v}^\delta \rangle - c \Delta \mathbf{v}^\delta, \end{aligned}$$

where the cross-term $-2\langle Dc, D\mathbf{v}^\delta \rangle$ arises from the product-rule identity $\Delta(c\mathbf{v}^\delta) = (\Delta c)\mathbf{v}^\delta + 2\langle Dc, D\mathbf{v}^\delta \rangle + c \Delta \mathbf{v}^\delta$. $\square$

Substituting these expressions and using the HJ PDE $\mathbf{v}_t^\delta = \frac{\delta}{2} \Delta \mathbf{v}^\delta - H$ yields the following residual equation.

Theorem 1 (Residual equation). *The transformed variable $\omega^\delta = e^{-c\mathbf{v}^\delta}$ satisfies*

$$\omega_t^\delta - \frac{\delta}{2} \Delta \omega^\delta = \omega^\delta \cdot R(t, \mathbf{x}), \quad (9)$$

where the **residual** is

$$R = -c_t \mathbf{v}^\delta + \delta \langle Dc, D\mathbf{v}^\delta \rangle (1 - c\mathbf{v}^\delta) - \frac{\delta}{2} (\mathbf{v}^\delta)^2 |Dc|^2 + \frac{\delta}{2} \mathbf{v}^\delta \Delta c. \quad (10)$$

The leading-order terms ($cH - \frac{\delta}{2}c^2|D\mathbf{v}^\delta|^2$) cancel by the choice of c ; only terms involving derivatives of c remain.

Proof. We set $f = -c\mathbf{v}^\delta$ so that $\omega^\delta = e^f$, and compute $\omega_t^\delta - \frac{\delta}{2}\Delta\omega^\delta$ in full. The derivation proceeds in six steps.

Step 1 (Factor out ω^δ). From (7) and (8):

$$\omega_t^\delta - \frac{\delta}{2} \Delta \omega^\delta = \omega^\delta \left[f_t - \frac{\delta}{2} (|Df|^2 + \Delta f) \right],$$

so the residual is $R = f_t - \frac{\delta}{2}|Df|^2 - \frac{\delta}{2}\Delta f$. We now expand each of these three pieces.

Step 2 (Compute f_t). Since $f = -c\mathbf{v}^\delta$ with $c = c(t, \mathbf{x})$:

$$f_t = -c_t \mathbf{v}^\delta - c \mathbf{v}_t^\delta.$$

Substitute the HJ PDE $\mathbf{v}_t^\delta = \frac{\delta}{2} \Delta \mathbf{v}^\delta - H$:

$$f_t = -c_t \mathbf{v}^\delta - c \left(\frac{\delta}{2} \Delta \mathbf{v}^\delta - H \right) = -c_t \mathbf{v}^\delta - \frac{c\delta}{2} \Delta \mathbf{v}^\delta + cH. \quad (11)$$

Step 3 (Compute $|Df|^2$). From $Df = -Dc\mathbf{v}^\delta - cD\mathbf{v}^\delta$, expand the square:

$$|Df|^2 = |Dc|^2 (\mathbf{v}^\delta)^2 + 2c\mathbf{v}^\delta \langle Dc, D\mathbf{v}^\delta \rangle + c^2 |D\mathbf{v}^\delta|^2. \quad (12)$$

Step 4 (Compute Δf). From the product-rule identity $\Delta(c\mathbf{v}^\delta) = (\Delta c)\mathbf{v}^\delta + 2\langle Dc, D\mathbf{v}^\delta \rangle + c\Delta\mathbf{v}^\delta$:

$$\Delta f = -\Delta(c\mathbf{v}^\delta) = -\Delta c\mathbf{v}^\delta - 2\langle Dc, D\mathbf{v}^\delta \rangle - c\Delta\mathbf{v}^\delta. \quad (13)$$

Step 5 (Assemble all terms). Substitute (11)–(13) into $R = f_t - \frac{\delta}{2}|Df|^2 - \frac{\delta}{2}\Delta f$:

$$\begin{aligned} R &= \underbrace{\left(-c_t \mathbf{v}^\delta - \frac{c\delta}{2} \Delta \mathbf{v}^\delta + cH \right)}_{f_t} \\ &\quad - \frac{\delta}{2} \underbrace{\left(|Dc|^2 (\mathbf{v}^\delta)^2 + 2c\mathbf{v}^\delta \langle Dc, D\mathbf{v}^\delta \rangle + c^2 |D\mathbf{v}^\delta|^2 \right)}_{|Df|^2} \\ &\quad - \frac{\delta}{2} \underbrace{\left(-\Delta c\mathbf{v}^\delta - 2\langle Dc, D\mathbf{v}^\delta \rangle - c\Delta\mathbf{v}^\delta \right)}_{\Delta f}. \end{aligned}$$

Distribute and list all seven resulting terms:

- (a) $-c_t \mathbf{v}^\delta$,
- (b) $-\frac{c\delta}{2} \Delta \mathbf{v}^\delta$,
- (c) $+cH$,
- (d) $-\frac{\delta}{2} c^2 |D\mathbf{v}^\delta|^2$,
- (e) $-c\delta \mathbf{v}^\delta \langle Dc, D\mathbf{v}^\delta \rangle$,
- (f) $-\frac{\delta}{2} |Dc|^2 (\mathbf{v}^\delta)^2$,
- (g) $+\delta \langle Dc, D\mathbf{v}^\delta \rangle$,
- (h) $+\frac{\delta}{2} \Delta c \mathbf{v}^\delta$,
- (i) $+\frac{c\delta}{2} \Delta \mathbf{v}^\delta$.

Step 6 (Cancellations and simplification).

(I) The $\Delta \mathbf{v}^\delta$ terms cancel. Terms (b) and (i): $-\frac{c\delta}{2} \Delta \mathbf{v}^\delta + \frac{c\delta}{2} \Delta \mathbf{v}^\delta = 0$.

(II) The leading-order terms cancel by the choice of c . From $c = \frac{2H}{\delta |D\mathbf{v}^\delta|^2}$, compute:

$$\frac{\delta}{2} c^2 |D\mathbf{v}^\delta|^2 = \frac{\delta}{2} \cdot \frac{4H^2}{\delta^2 |D\mathbf{v}^\delta|^4} \cdot |D\mathbf{v}^\delta|^2 = \frac{2H^2}{\delta |D\mathbf{v}^\delta|^2} = cH.$$

Therefore terms (c) and (d): $cH - \frac{\delta}{2} c^2 |D\mathbf{v}^\delta|^2 = 0$.

(III) Factor the $\langle Dc, D\mathbf{v}^\delta \rangle$ terms. Terms (e) and (g):

$$\delta \langle Dc, D\mathbf{v}^\delta \rangle - c\delta \mathbf{v}^\delta \langle Dc, D\mathbf{v}^\delta \rangle = \delta \langle Dc, D\mathbf{v}^\delta \rangle (1 - c \mathbf{v}^\delta).$$

(IV) Collect the surviving terms: (a), (III), (f), (h):

$$R = -c_t \mathbf{v}^\delta + \delta \langle Dc, D\mathbf{v}^\delta \rangle (1 - c \mathbf{v}^\delta) - \frac{\delta}{2} (\mathbf{v}^\delta)^2 |Dc|^2 + \frac{\delta}{2} \mathbf{v}^\delta \Delta c.$$

Every surviving term involves a derivative of c (c_t , Dc , or Δc). When c is constant these all vanish and $R \equiv 0$, recovering the exact Cole-Hopf result. $\square$

Corollary 1. $R \equiv 0$ if and only if c is constant, which holds when $H/|D\mathbf{v}^\delta|^2$ is independent of $(\mathbf{x}, t)$. For $H = \frac{1}{2} |\mathbf{p}|^2$, $c = 1/\delta = \text{const}$, recovering Proposition 1.

Remark 1 (Errata in the ICML submission). (i) Proposition 3.1 in the submission states the linearized equation as $\partial_t \omega^\delta - \frac{\delta}{2} \Delta \omega^\delta = H$; the RHS should be 0 (for the exact case) or $\omega^\delta R$ (for the general case). (ii) Equations (11) and (12) use $|Dg|^2/H^\delta$ as a prefactor; this equals $|D\mathbf{v}^\delta|^2/H$ only at $t = 0$. For $t > 0$, the correct prefactor involves $|D\mathbf{v}^\delta|^2$ and $H(t; \mathbf{x}, D\mathbf{v}^\delta)$ at the current time. (iii) Lemmas 3.3 and 3.4 are formally correct as identities but are not self-contained: they involve $D^2 \mathbf{v}^\delta$, $D_t D\mathbf{v}^\delta$, DH , and $D_t H$, which are themselves unknown.

3 Quasi-Linearization Algorithm

For general H , we propose an iterative scheme:

Algorithm (Quasi-Linearized Cole-Hopf).

1. **Initialize:** $\mathbf{v}^{(0)}(t; \mathbf{x}) = g(\mathbf{x})$, $c^{(0)} = \frac{2H(t; \mathbf{x}, Dg)}{\delta |Dg|^2}$.
2. **For** $k = 0, 1, 2, \dots$:
 - (a) Freeze $c^{(k)}$ at the current iterate.
 - (b) Solve the heat equation $\omega_t = \frac{\delta}{2} \Delta \omega$ with initial data $\omega^{(k)}(0; \mathbf{x}) = e^{-c^{(k)}g(\mathbf{x})}$.
 - (c) Recover $\mathbf{v}^{(k+1)} = -(1/c^{(k)}) \log \omega^{(k+1)}$.
 - (d) Update $D\mathbf{v}^{(k+1)}$ and $c^{(k+1)} = \frac{2H(t; \mathbf{x}, D\mathbf{v}^{(k+1)})}{\delta |D\mathbf{v}^{(k+1)}|^2}$.
 - (e) Check convergence: $\|\mathbf{v}^{(k+1)} - \mathbf{v}^{(k)}\| / \|\mathbf{v}^{(k)}\| < \varepsilon$.

4 Gaussian Expectation Formulas

At each iteration, the heat equation is solved via the Green's function:

Lemma 1 (Value function recovery). *Given the heat equation with initial data $\omega(0; \mathbf{x}) = e^{-c g(\mathbf{x})}$, the value function at time t is*

$$\mathbf{v}^\delta(t; \mathbf{x}) = -\frac{1}{c} \log \mathbb{E}_{\mathbf{z} \sim \mathcal{N}(0, I_n)} [e^{-c g(\mathbf{x} + \sigma \mathbf{z})}], \quad \sigma = \sqrt{\delta(T - t)}. \quad (14)$$

For numerical stability, use the log-sum-exp identity: $-\frac{1}{c} \log \frac{1}{J} \sum_{j=1}^J e^{-c g(\mathbf{x} + \sigma \mathbf{z}_j)}$ with $\mathbf{z}_j \stackrel{iid}{\sim} \mathcal{N}(0, I_n)$.

Lemma 2 (Gradient recovery). *The spatial gradient is recovered as an importance-weighted ratio:*

$$D\mathbf{v}^\delta(t; \mathbf{x}) = \frac{\mathbb{E}_{\mathbf{z}} [Dg(\mathbf{x} + \sigma \mathbf{z}) e^{-c g(\mathbf{x} + \sigma \mathbf{z})}]}{\mathbb{E}_{\mathbf{z}} [e^{-c g(\mathbf{x} + \sigma \mathbf{z})}]}. \quad (15)$$

Proof. From $\omega = e^{-c\mathbf{v}}$ and the heat-kernel representation $\omega(t; \mathbf{x}) = \mathbb{E}[e^{-c g(\mathbf{x} + \sigma \mathbf{z})}]$:

$$D\omega = \mathbb{E}[-c Dg(\mathbf{y}) e^{-c g(\mathbf{y})}] \cdot \frac{1}{\sigma} \frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \mathbb{E}[Dg(\mathbf{y}) \cdot (-c) e^{-c g(\mathbf{y})}].$$

Wait — more carefully: $D_{\mathbf{x}} \omega(t; \mathbf{x}) = \mathbb{E}[D_{\mathbf{x}} e^{-c g(\mathbf{x} + \sigma \mathbf{z})}] = -c \mathbb{E}[Dg(\mathbf{y}) e^{-c g(\mathbf{y})}]$. Since $\omega = e^{-c\mathbf{v}}$ and $D\omega = -c e^{-c\mathbf{v}} D\mathbf{v} = -c\omega D\mathbf{v}$, we get $D\mathbf{v} = D\omega/(-c\omega)$, which gives (15). $\square$

5 Error Analysis

5.1 Quasi-linearization error

The residual R in (10) involves Dc , Δc , c_t . Since $c = 2H/(\delta |D\mathbf{v}|^2)$, we have $Dc = O(D^2\mathbf{v}, DH)$. For smooth solutions with bounded Hessian $\|D^2\mathbf{v}\| \leq M_2$:

$$\|R\|_\infty \leq C(\delta, M_2, \|DH\|, \|\mathbf{v}\|_\infty) \cdot \|Dc\|_\infty.$$

When the iteration converges ($\mathbf{v}^{(k+1)} \approx \mathbf{v}^{(k)}$), the frozen coefficient $c^{(k)}$ stabilizes and $R \rightarrow 0$ effectively.

5.2 Monte Carlo error

The standard MC estimator for (14) converges at rate $O(1/\sqrt{J})$ where J is the number of samples. The effective variance depends on c and the range of g : when $c\|g\|_\infty \gg 1$ (small δ), the exponential weights become concentrated and the effective sample size shrinks. Importance sampling with a tilted proposal (e.g., Laplace approximation around the mode of e^{-cg}) can reduce variance.

5.3 Viscosity approximation error

By Crandall-Lions (1984), $\|\mathbf{v}^\delta - \mathbf{v}\|_\infty \leq k\sqrt{\delta}$ where $\mathbf{v}$ is the inviscid viscosity solution. Thus smaller δ improves the approximation quality but increases the MC variance (the weights become more peaked). This creates a fundamental bias-variance tradeoff controlled by δ .